W23-OIT-367-02 – Business Intelligence from Big Data
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SIGNAL VS. NOISE IN PREDICTING FUTURE VENTURE CAPITAL FUNDRAISING SUCCESS

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EXECUTIVE SUMMARY

Venture capital is a form of early-stage equity-based financing for startup companies that have high potential for long-term growth. With the average venture-backed company burning cash and thus requiring three to four rounds of investment financing prior to an eventual "exit" event, venture-backed companies are generally evaluated on their ability to hit certain milestones (e.g., revenue growth, users, etc.) within ~12-24-month periods in order to attract further venture capital investment that will be used to invest in growth. This frequent dependence on the willingness of a *future* venture investor to back the company at the next stage of fundraising makes it critical for any venture capitalist to evaluate the likelihood that a business they invest in will be able to raise the next round of funding. Considering that, our project sought to answer the question of "what characteristics of a company and its fundraising history determine venture-backed company success?", defining a given financing round as *successful* if the company is able to raise a future round of venture capital funding.

Our analysis suggests the presence of a strong positive signaling effect in venture capital investing (i.e., when the presence of one respected venture capital investor on a company's cap table is perceived as a positive "signal" and increases the likelihood that other venture capital investors will want to invest), both in terms of the composition of the quality of investors on the capital table and the amount of cumulative venture capital funding raised to-date. However, we also identify two likely major flaws in the analysis: i) the nature of our analysis involves sampling on the dependent variable (the data does not include the broad set of companies that have been unable to fundraise and thus reflect the counterfactual of actual startup fundraising failure), and ii) the vast majority of the data we were given reflects trends in venture capital financing during a period of low interest rates, low inflation, economic recovery, and strong overall growth in venture capital as an asset class. While we find our results interesting and partially explanatory (though still flawed) on historical trends in venture capital, we acknowledge that without a more expansive set of data (as described above), our analysis is not predictive of the success of a company that approaches ThomVest Ventures for funding, and therefore, should not be used to make investment decisions.

1 INTRODUCTION

Venture capital is a form of early-stage equity-based financing for startup companies that have high potential for long-term growth. With the average venture-backed company burning cash and thus requiring three to four rounds of investment financing prior to an eventual "exit" event such as an initial public offering (IPO) or an acquisition, startup funding rounds are typically structured in stages. These stages range from what is often called a seed round, meant to supply startups with the capital they need to build a business foundation, to series A / B / C rounds once a company has found product-market fit (a good market with a product capable of satisfying that market, thus shifting the focus to investments that support growth).

Given a traditional venture-backed company's dependence on multiple rounds of financing prior to a successful exit, venture-backed companies are generally evaluated on their ability to hit certain milestones (e.g., revenue growth, users, etc.) within ~12-24-month periods in order to attract further venture capital investment that will be used to invest in growth. Though the success of a venture capital investment is ultimately judged based on the amount of capital returned at exit, this frequent dependence on the willingness of a *future* venture investor to back the company at the next stage of fundraising makes it critical for any venture capitalist to evaluate the likelihood that a business they invest in will be able to raise the next round of funding.

With the importance of the problem understood, our project answers the question of "what characteristics of a company and its fundraising history determine venture-backed company success?". We specifically define a given financing round as *successful* if the company is able to raise a future round of venture capital funding. Beyond evaluating the impact of company & current / former fundraising details, we also specifically examine the role that both positive and negative signaling (generally defined as the presence and departure, respectively, of a respected venture capital investor from a company's fundraising history) play in financing round success.

2 DATA

The majority of the data we used for this project was provided directly by our client, Thomvest Ventures, but is publicly available for purchase. Thomvest provided us with four files: "companies.csv", "funding.csv", "investors.csv", and "harmonic_enrichments.json". The first of these two files included data sourced from PitchBook (https://pitchbook.com/) and Crunchbase (https://www.crunchbase.com/). The raw company data included 1,033 rows representing specific U.S.-based venture-backed startups having raised a series A or B in the last two years and included 30 columns of company attributes (e.g., company name, website, description, founding date, location, etc., amongst others). The raw funding data included 5,752 rows representing individual capital raises from the companies included in the raw company data and included 24 columns of funding attributes (e.g., round, number of investors, announcement date, amount raised, etc., amongst others). The raw investor data included 382 rows representing each investor referenced in the funding data along with Thomvest's internal tiering (i.e., reputational classification) of the investor. 57 of the investors are listed are counted as tier 1 (e.g., traditional blue-chip firms like Sequoia and Andreessen-Horowitz) while 275 are counted as tier 2 and 50 are counted as tier 3. Lastly, the harmonic enrichment data is sourced from Harmonic.ai (https://www.harmonic.ai/) and includes additional company and funding attribute data as well as traction metrics (e.g., Facebook follower count) in a JSON file. In addition to the data provided by Thomvest (which we both incorporate directly and transform to create our predictor variables, discussed below), we also add two new variables T10 (a numerical variable representing the market 10-Year Treasury Yield on the date of fundraising announcement, a proxy for liquidity and the macroeconomic environment) and is_english_word (a binary variable representing whether or not a given startup's name is a word in the English dictionary).

Next, having aggregated our entire data set, we performed the data clean-up and preprocessing to prepare our data for classification analysis. The most critical steps (non-exhaustive) we undertook in cleaning and preprocessing the data were the following:

- We dropped all columns that contained NaN in every cell.

- We transformed the raw data we were provided with into the specific variables we intended to use in our classification analysis. A simple definition of how each specific variable was transformed is included in Exhibit 1 below and can be found in more detail in our code submission.
- We converted our categorical variables into dummy / indicator variables such that we can run our regression and additional analyses on numeric form data.
- Using BERT sentence transformation and K-means clustering, we converted 769 different industry tags into nine different industry clusters.
- We removed all of the rows for which the company name is NaN due to lack of response variable.
- We removed all of the rows for which the current round represents the company's *last* round of fundraising (due to lack of response variable).
- We merged all of our data sources into a single DataFrame / clean file with each row representing an individual fundraising round at a given company (and with prior round details included in-row that we can use for analysis).

It is worth highlighting that we were unable to utilize a considerable portion of the data we received, encompassing aspects such as social media trends (e.g., followers on Facebook, LinkedIn, Twitter, PitchBook, etc.), employee count, and post-money valuations of funding rounds, amongst others. This limitation arose from the scarcity of data points for certain metrics (e.g., post-money valuations) and the fact that some data was restricted to only the previous year (e.g., social media trends) or a single point in time (e.g., employee count).

Upon aggregation of our single DataFrame / clean file, the last step we took was to define our intended response variables and add them as columns. We finalized our data set by doing this and adding response variables successTierOne (a binary variable for if a given current round row successfully raised a future round including a tier "1" venture investor), successTierOneTwo (same, but with a tier "1" or "2" venture investor), and successTierOneTwoThree (same, but with a tier "1", "2", or "3" venture investor). Overall, we are left with 3,287 remaining rows upon which we can begin performing our classification analysis.

EXHIBIT 1: PREDICTOR VARIABLE SOURCES AND DESCRIPTIONS

Variable	Source	Definition
days_from_last_round	Transformation of Thomvest data	**Numerical:** Days between current and prior round announcement
log_money_raised_last_round	Transformation of Thomvest data	**Numerical:** Natural logarithm of funding raised last round
log_money_raised_this_round	Transformation of Thomvest data	**Numerical:** Natural logarithm of funding raised current round
top_tier_abandonement	Transformation of Thomvest data	**Binary:** Current rounds without a Tier 1 investor that had a Tier 1 investor in prior rounds
is_silicon_valley	Transformation of Thomvest data	**Binary:** Company location listed as San Francisco or Palo Alto
cluster_industry_tags_x	Transformation of Thomvest data	**Categorical:** 9 industry clusters assigned using K-means
year_founded_x	Thomvest raw data	**Categorical:** 25 different years in which companies were founded
year_round_announced_x	Thomvest raw data	**Categorical:** 25 different years in which rounds were announced
month_round_announced_x	Thomvest raw data	**Categorical:** 12 different months in which rounds were announced
day_round_announced_x	Thomvest raw data	**Categorical:** 31 different days (of the month) on which rounds were announced
day_of_the_week_announced_x	Thomvest raw data	**Categorical:** 7 different days (of the week) on which rounds were announced
highest_tier_1	Transformation of Thomvest data	**Binary:** Investor tier of highest-tiered investor is a "1" for current round
highest_tier_2	Transformation of Thomvest data	**Binary:** Investor tier of highest-tiered investor is a "2" for current round
highest_tier_3	Transformation of Thomvest data	**Binary:** Investor tier of highest-tiered investor is a "3" for current round
highest_tier_4	Transformation of Thomvest data	**Binary:** No investors in tiers "1", "2", or "3" for current round
B2C	Thomvest raw data	**Binary:** Company sells to consumer customers
B2B_B2C	Thomvest raw data	**Binary:** Company sells to business and consumer customers
B2B	Thomvest raw data	**Binary:** Company sells to business customers
has_linkedin_url	Transformation of Thomvest data	**Binary:** Company has a LinkedIn URL
has_facebook_url	Transformation of Thomvest data	**Binary:** Company has a Facebook URL
has_twitter_url	Transformation of Thomvest data	**Binary:** Company has a Twitter URL
investor_count	Thomvest raw data	**Numerical:** Total number of investors in current round
name_length	Transformation of Thomvest data	**Numerical:** Character length of company's name
url_length	Transformation of Thomvest data	**Numerical:** Character length of company's website URL
T10	New data	**Numerical:** Market 10-Year Treasury Yield on date of round announcement
is_english_word	New data	**Binary:** Startup's name is a word in the English dictionary
Tier1HighestPriorRound	Transformation of Thomvest data	**Binary:** Investor tier of highest-tiered investor is a "1" for prior rounds
Tier2HighestPriorRound	Transformation of Thomvest data	**Binary:** Investor tier of highest-tiered investor is a "2" for prior rounds
Tier3HighestPriorRound	Transformation of Thomvest data	**Binary:** Investor tier of highest-tiered investor is a "3" for prior rounds
series_x	Thomvest raw data	**Categorical:** 5 different types of rounds representing Series A, B, C, D, and E

3 METHODS

After we collected the data, we then randomized and partitioned our data set, using 70% as the training set and the remaining 30% as the test set. We ran each type of classification analysis on all three selected response variables, each producing results that are applicable depending on whether our client wants to focus on new rounds from only the most top-tier venture firms or whether the more relevant analysis considers a broader potential set of investors. In order to decide which model was the most useful model, we took the position of Thomvest Ventures. In order to obtain an allocation in a startup's cap table, funds that prefer a minority stake must often verbally commit to provide funding before the startup has found a lead investor. It then hopes that a top tier investor will invest in later rounds to help propel the company to success. For the purposes of focusing on a "primary" model and given our client's key question of how we predict company success, defining success as raising a new round of venture capital, ideally from a top-tier venture firm, we define successTierOneTwo as our primary response variable in our classification analysis.

Next, we ran four types of classification analyses on our selected response variable: i) a logistic regression using each predictor highlighted in Exhibit 1, ii) a classification tree with depth 3 (for simple visualization), iii) a random forest, and iv) a gradient boosting machine. For each of these analyses, we evaluated our classification performance using the average areas under the curve[1] (AUCs) and standard errors generated in 5-fold cross-validation[2]. As shown in the AUC display table in Exhibit 2, we can clearly see that while the standard errors only have modest differences, the AUC differences are significant, and it is clear that Random Forest > Gradient Boosting Machine > Classification Tree > Logistic Regression in terms of classification performance. Due in part to being the highest performance model but also due to the specific disadvantages of both logistic regression and classification tree analyses (for logistic regression, variable selection is ultimately quite time-intensive, our chosen predictor variables most likely demand nonlinear and interaction terms that are more effectively captured in a tree-based approach, and logistic regression lacks the transparency of a tree-based approach; for classification tree, the splits are chosen greedily and do not explore all possible relationships between predictors—this weakness is eliminated in the random forest analysis and shown to be meaningful given the large AUC gap between the two), we consider the random forest analysis to represent our final prediction model. The primary downside of the random forest analysis relative to a simpler classification tree analysis is that the interpretability of a simple decision tree is lost. For this reason, and as discussed further in our Results section, we also leverage the classification tree analysis given its potential practicality for our client.

Lastly, upon seeing the impact of signaling in visualization of both our classification trees and random forest feature importance analysis from above, we elected to re-run the same analyses highlighted in the paragraph above *exclusive* of the current round tiering data (and thus independently of the signaling effect). We discuss our interpretation further in Results but also include an AUC display table for these analyses in Exhibit 3.

4 RESULTS

Though not our "final model" for reasons discussed in our Methods section, we look first to the interpretation of our classification tree analysis in Exhibits 4 and 5. In Exhibit 4 (test set AUC of 0.69), we observe quickly that the four most predictive variables in the classification tree are highest_tier_2, highest_tier_1, log_money_raised_last_round, and log_money_raised_this_round. Intuitive explanations for this might be i) that there are significant dollars looking to follow well-respected tier 1 and tier 2 venture capitalist investors, and ii) raising a significant amount of funding ultimately pushes a startup's timeline to fail back and de-risks the near-term, naturally leading to greater success in future fundraising. We also recognize this trend could be an artifact of the

[1] A common measure of classification test accuracy, AUC tells the degree of separability / how much a given model is capable of distinguishing between classes. AUC of 0.5 is the worst possible model one can build and AUC of 1 is a perfect classification model.

[2] In 5-fold cross-validation, resampling enabled us to repeat the training and validation process five times, each time using a slightly different part of the data for training and validation. The output of this cross-validation is that we get mean and variance of performance instead of a single point and thus we can with higher certainty say that one approach to classification is better than the other (and ensure that a given model is not overfitted).

chosen response variable and top tier funds are also likely to provide follow on investments in future rounds. In Exhibit 5 (test set AUC of 0.66), which excludes the investor tier binary variables (and thus doesn't explicitly capture investor signaling in the current round that ThomVest may not have access to when making a decision), we observe different most predictive variables including in silicon_valley, name_length, and url_length. Intuitive explanations for this might be that if one cannot simply follow the signal put out by top-tier investors, one might be best off in pursuit of companies located in Silicon Valley with short company names and URLs. Though likely not causal, these variables might represent a proxy for marketing savvy leadership, companies who have raised enough capital to buy the best domains, or companies with the confidence and funding base to build in San Francisco.

Looking now at the random forest analysis we consider our final model, we look to the interpretation of the measured importance by feature in Exhibits 6 and 7. In Exhibit 6 (test set AUC of 0.83), we see consistent with Exhibit 4 that investor tier and balance sheet have a significant impact on success classification. In addition, in Exhibit 7 (test set AUC of 0.81), we see that some of the most important predictor variables when signal is excluded are consistent with what we saw in the classification tree.

In stepping back and considering both the statistical significance and the generalizability of the results presented, we would posit that the results are likely generalizable to future Pitchbook, Crunchbase, and Harmonic datasets, but are likely not helpful in assisting ThomVest Ventures make investment decisions given what we see as a few flaws in the data. First, we are confident that the findings are generalizable given that the 5-fold cross-validation had such a low variation across a large dataset. However, a few of the flaws that have surfaced include i) sampling on the dependent variable (the client data set we are working with is based largely on companies that have been able to recently successfully fundraise and thus don't reflect the counterfactual of actual startup businesses that have attempted to fundraise and been unsuccessful), and ii) though meaningful in size of the data set, the significant majority of data included is recorded within a specific macroeconomic time period where low interest rates and economic recovery paved the way for valuation increases and increased deployment of capital into the asset category. These two flaws make it unlikely that the conclusions can be used for investment decisions.

5 CONCLUSIONS

To summarize, in undertaking this project, we sought to help our client answer the question of "what characteristics of a company and its fundraising history determine venture-backed company success?", defining a given financing round as *successful* if the company is able to raise a future round of venture capital funding. Our analysis suggests the presence of a strong positive signaling effect in venture capital investing (i.e., when the presence of one respected venture capital investor on a company's cap table is perceived as a positive "signal" and increases the likelihood that other venture capital investors will want to invest), both in terms of the composition of the quality of investors on the capital table and the amount of cumulative venture capital funding raised to-date. However, we also identify two likely major flaws in the analysis: i) the nature of our analysis involves sampling on the dependent variable (the data does not include the broad set of companies that have been unable to fundraise and thus reflect the counterfactual of actual startup fundraising failure), and ii) the vast majority of the data we were given reflects trends in venture capital financing during a period of low interest rates, low inflation, economic recovery, and strong overall growth in venture capital as an asset class. While we find our results interesting and partially explanatory (though still flawed) on historical trends in venture capital, we acknowledge that without a more expansive set of data (as described above), our analysis is not predictive of the success of a company that approaches ThomVest Ventures for funding, and therefore, should not be used to make investment decisions.

6 APPENDIX

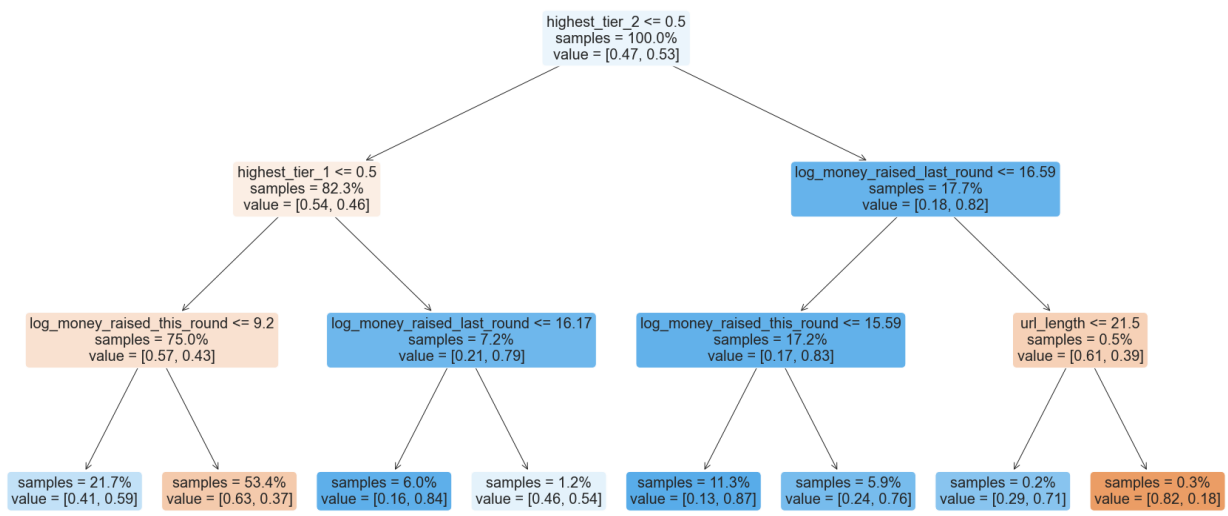
EXHIBIT 2: AUC DISPLAY TABLE FOR RESPONSE VARIABLE successTierOneTwo

Statistics	Logistic Regression	Tree	Random Forest	Gradient Boosted Machine
Average AUC	0.6546	0.6939	0.8335	0.8084
Standard Error	0.0122	0.0041	0.0045	0.0048

EXHIBIT 3: AUC DISPLAY TABLE FOR RESPONSE VARIABLE successTierOneTwo (EXCLUDING CURRENT ROUND TIERING DATA)

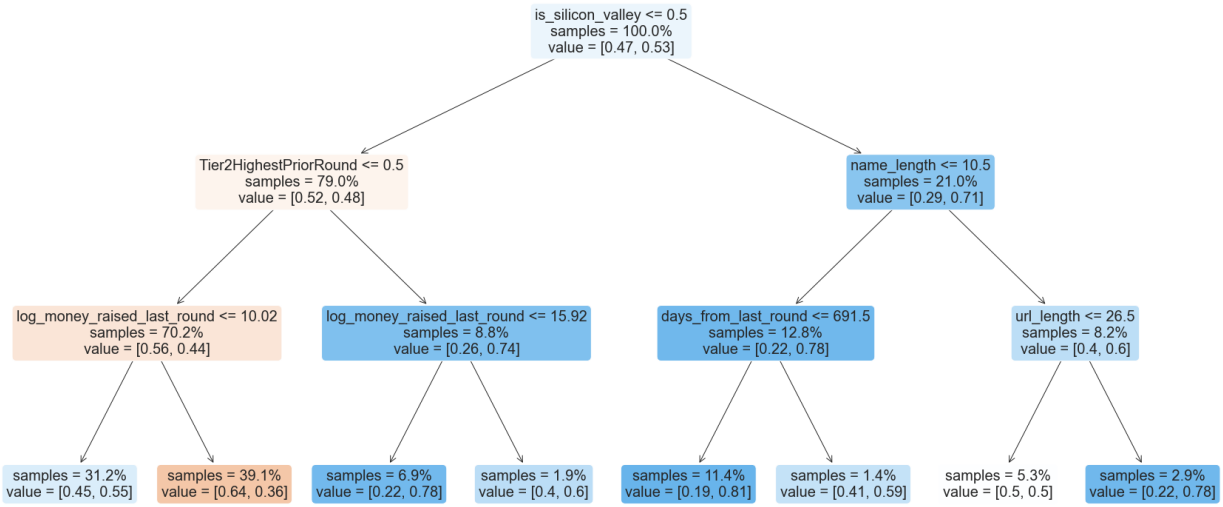
Statistics	Logistic Regression	Tree	Random Forest	Gradient Boosted Machine
Average AUC	0.6047	0.6584	0.8198	0.786
Standard Error	0.0088	0.0041	0.0031	0.0029

EXHIBIT 4: DECISION TREE OF DEPTH 3 ON RESPONSE VARIABLE successTierOneTwo



The top box contains 100% of the samples ("samples=100%"), meaning all of the training data starts at this node. It also shows that the means in the data available at this node fraction of 0's and 1's are 0.47 and 0.53 respectively ("value = [0.47, 0.53]"). Equivalently, this means the probability of "SuccessTierOneTwo = 1" at this node is 53%. "highest_tier_2<=0.5" means after this node, we split the data, by looking at the variable "highest_tier_2". All samples that satisfy the condition, will move to the left, otherwise, will move the right. Therefore, the condition splits the data. Looking at the bottom left, the vast majority of rows (75.0%) only pass the conditions of what is the tier of the top investor is from the current round and the natural logarithm of how much money is raised this round.

EXHIBIT 5: DECISION TREE OF DEPTH 3 ON RESPONSE VARIABLE successTierOneTwo (EXCLUDING CURRENT ROUND TIERING DATA)



See the exhibit above for an explanation of classification tree traversal. Looking at the bottom left, the vast majority of rows (70.2%) only pass the conditions of what was the tier of the top investor from the previous round, how much money was raised, and if the company is from Silicon Valley.

EXHIBIT 6: BAR GRAPH OF RANDOM FOREST FEATURE IMPORTANCE ON RESPONSE VARIABLE successTierOneTwo

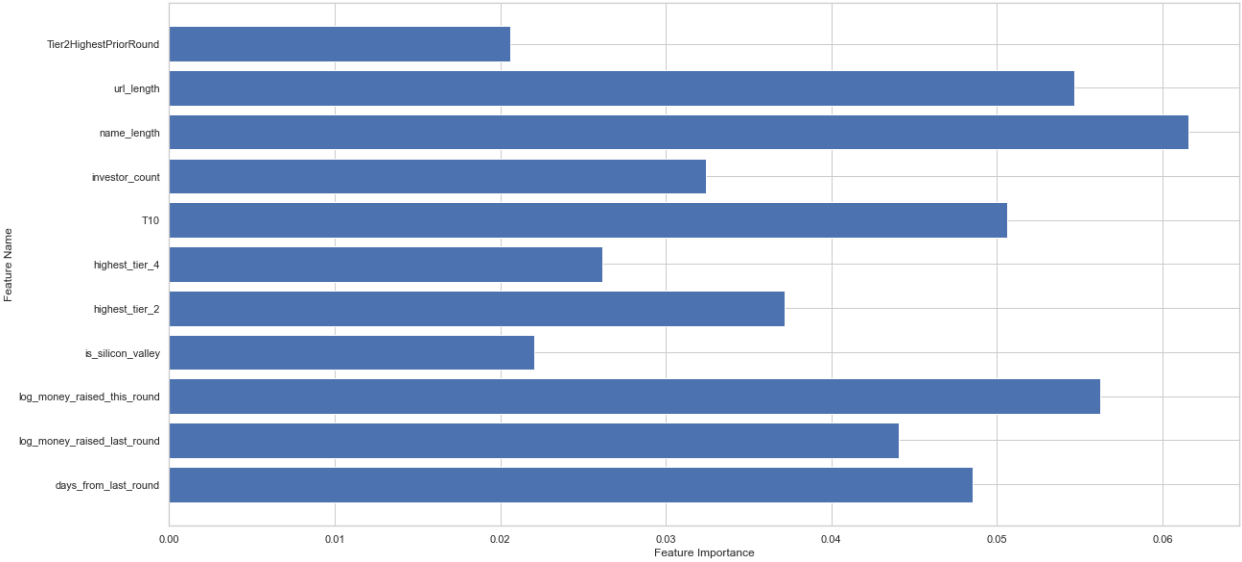
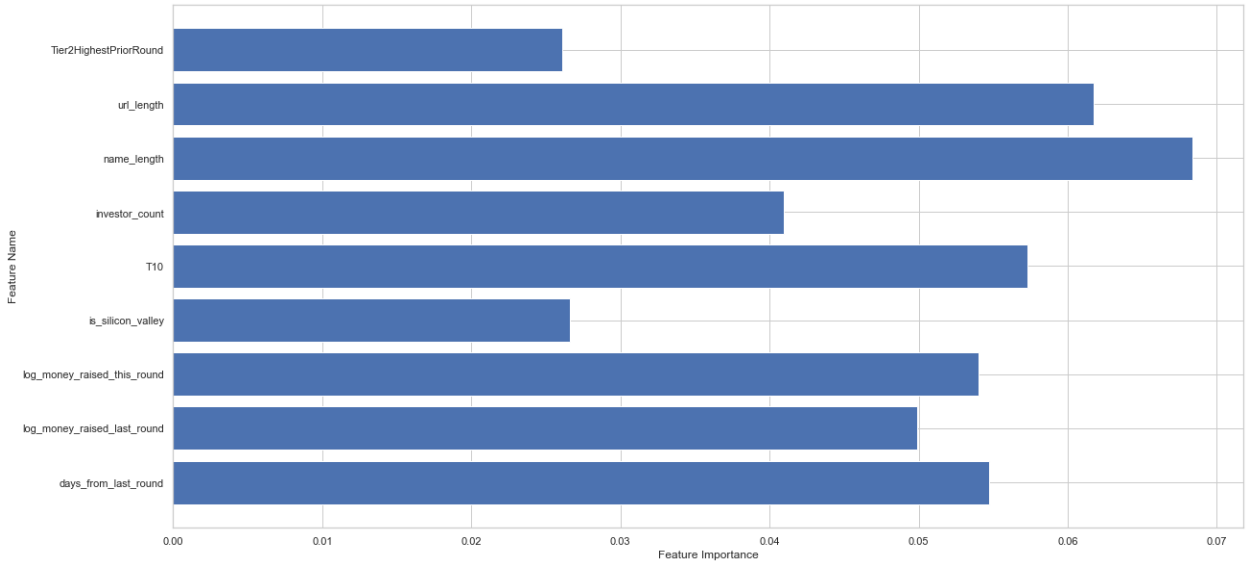



EXHIBIT 7: BAR GRAPH OF RANDOM FOREST FEATURE IMPORTANCE ON RESPONSE VARIABLE successTierOneTwo (EXCLUDING CURRENT ROUND TIERING DATA)